Joon-Hwi Kim

Ph.D., Physics

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Profile

Theoretical and mathematical physicist with 20+ papers, 200+ citations, and 5+ invited talks globally, with strong backgrounds in quantum field theory, perturbation and scattering theory, differential and symplectic geometry, and symbolic computation methods. Skilled in translating abstract mathematical structures into calculational frameworks using Mathematica or Python.

Education

Ph.D. in Physics, Caltech (Pasadena, CA, USA) [Advisor: Clifford Cheung]	09/2021-06/2026
B.S. in Physics, Seoul National University (Seoul, Korea) [Advisor: Sangmin Lee]	02/2017-02/2021
Seoul Science High School for the Gifted (Seoul, Korea)	02/2014-02/2017

Awards and Honors

Gold Medal in the 46th International Physics Olympiad (Tata Institute of Fundamental Research)	07/2015
Gold Prize in the 5th Hanwha Science Challenge (₩20,000,000, Hanwha Co., Ltd.)	08/2015
Presidential Science Scholarship, Korea Student Aid Foundation (Full tuition)	03/2017-02/2021
Korea Talent Award (Korea Foundation for Advancement of Science & Technology, 50 individuals/nation)	11/2016
Research Grant, Ilju Academy of Science of Culture ($120,000)	08/2021-07/2025

Skills

Programming	Python, Mathematica, Symbolic Tensor Manipulation via xAct, Natural Language Processing, JavaScript, HTML
Physics	Quantum Mechanics and Field Theory, Statistical Mechanics and Field Theory, Perturbation Theory, Path Integral Methods, Strong-Field Gravity and Black Holes
Mathematics	Differential/Symplectic Geometry, Complex Analysis, Lie Groups/Algebras, Homology/Cohomology, Category Theory, Twistors/Spinors
Miscellaneous	Adobe Illustrator, Adobe Photoshop, Rhinoceros 3D, MIDI Sequencers, LaTeX
Languages	Korean (Native), English (Fluent)

Selected Publications

[2412.19611]	Newman-Janis Algorithm from Taub-NUT Instantons. **JHK**. *Physical Review Letters* 136 (2026) 23, 231401	15 citations
[2403.01837]	Generalized Symmetry in Dynamical Gravity. C. Cheung, M. Derda, **JHK**, V. Nevoa, I. Rothstein, N. Shah. *JHEP* 10 (2024) 007	28 citations
[2405.17056]	Massive Twistor Worldline in Electromagnetic Fields. **JHK**, J.-W. Kim, S. Lee. *JHEP* 8 (2024) 080	31 citations
[2410.22988]	Classical Eikonal from Magnus Expansion. **JHK**, J.-W. Kim, S. Kim, S. Lee.	25 citations

JHEP 1 (2025) 111

[2405.09518] Single Kerr-Schild Metric for Taub-NUT Instanton. **JHK**. 23 citations
Physical Review D 111 (2025)

[2509.23600] Geodesic Deviation to All Orders via a Tangent Bundle Formalism. **JHK**. 6 citations
General Relativity and Gravitation 58 (2026) 7 75 (Mathematica codes available)

[2509.23600] The Diagrammar of Quantum Magnusian.
L. Guo, **JHK**, J.-W. Kim, S. Kim, S. Lee, J.-R. Li. (Mathematica codes available)

Selected Talks

Building Black Holes from Taub-NUT Instantons — 02/05/2025
Self-Force Group Seminar Series, *University of Southampton (Hampshire, UK)*

Newman-Janis Shift and Beyond — 01/15/2025
Séminaire Amplitudes et Gravitation sur l'Yvette, *IHÉS/IPhT (Bures-sur-Yvette, France)*

One-Form Symmetry in Dynamical Gravity — 04/18/2024
Theoretical Elementary Particle (TEP) Physics Seminar, *UCLA (Los Angeles, California, USA)*

Twistor Particle Programme Rebooted: A Zig-Zag Theory of Massive Spinning Particles — 11/17/2023
QFT-Relativity Seminars, *Oxford University (Oxford, UK)*

Symplectic Perturbation Theory — 03/12/2023
California Amplitudes Meeting 2023, *UCSD (San Diego, California, USA)*

Selected Academic Travels and Activities

Mathematical Structures in Scattering Amplitudes, *Galileo Galilei Institute for Theoretical Physics (Florence, Italy)*; Invitation by Lionel Mason	08/2024
Symmetries and Anomalies: a Modern Take, *Institut des Hautes Études Scientifiques (Bures-sur-Yvette, France)*	06-07/2024
Amplitudes 2024, *Institute of Advanced Study (Princeton, New Jersey, USA)*	06/2024
Summer School on Numerical Relativity: Black Hole Imaging with Supercomputer Assistance, *Asian-Pacific Center for Theoretical Physics (Seoul, Korea)*	06/2019

Selected Coursework

Physics	String Theory (Caltech), Quantum Field Theory (Caltech, SNU), Knot Theory and Quantum Topology (Caltech), Advanced Quantum Field Theory and String Theory (KAIST/KIAS), Topological Field Theory (Caltech), Properties of Solids (SNU), Electronics and Precision Measurement (SNU), Fluid Dynamics (SNU), General Relativity (Caltech, SNU), Statistical Mechanics (SNU), Nuclear and Particle Physics (SNU)
Mathematics	Advanced Mathematical Methods of Physics (Caltech), Topics in Mathematical Physics (Caltech), Riemannian Geometry (Caltech), Theory of Lie Groups (SNU)
Programming and Data Processing	Computational Linguistics and Natural Language Processing (SNU), Cognitive Linguistics (SNU), Music Perception and Cognition (SNU), Perception of Visual Art (SNU), Numerical Physics: Artificial Intelligence for Physics (SNU)

Media

How to turn the complex mathematics of vector calculus into simple pictures. *MIT Technology Review*, Nov 2019.

Teaching

- Caltech (2021-): Teaching Assistant for Quantum Mechanics, Mathematical Methods of Physics, Theoretical Astroparticle Physics, General Relativity, Electromagnetism, Classical Mechanics, and related courses.
- Korean Physics Society and Seoul National University (2017-2019): Lecturer and Teaching Assistant roles in the summer and winter intensive training programs for the International Physics Olympiad.
- Seoul Science High School for the Gifted: Lecturer (2017)
  - "Physics and the Imagination": 10 lectures on mathematical physics following Schutz'80 and Nakahara'03.
  - "Escher's Studio": Directed a one-year interdisciplinary program investigating the intersections between mathematics (group theory, algebraic topology, complex analysis, conformal mapping), physics (symmetry, gauge theory), and computer science (Javascript programming, generative art, dynamic graphic design).